ARCHITECTURE AND DECISION RECORD

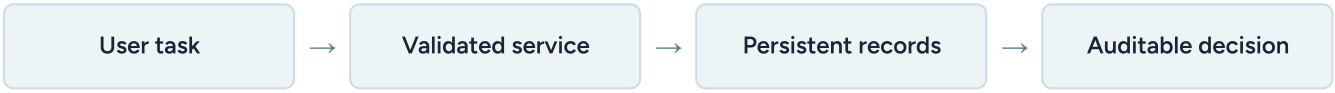
A library circulation and resource reservation portal

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Laravel | SQL | Data modeling | Reservation workflows

Problem and contribution

This original design case study explores how to maintain consistent copy-level availability across reservation, issue, return, renewal, and inventory workflows. I authored the requirements breakdown, architecture boundaries, data relationships, failure scenarios, and review rationale presented here. The design is a fictional presentation example, with no claim of a production deployment or measured client result.



Architecture decisions and tradeoffs

- Use copy identifiers to prevent two borrowers from receiving the same physical item.
- Represent renewal as a dated event with eligibility checks so an earlier due date remains auditable.
- Do not infer overdue fees or student discipline; the demonstration records due dates and staff decisions only.

Core records

Record	Responsibility
users	A borrower has many reservations and loans; staff authorization is independent.
titles	A title has many physical copies, each with a unique accession code.
copies	A copy can have loan history but only one active issue.
reservations	Reservations target a title; allocation identifies the eventual copy.

The service layer owns authorization and business constraints. Interface validation improves the experience but cannot replace server checks. Event records preserve attributable decisions and their timing without logging unnecessary private content.

VERIFICATION AND REVIEW PRACTICE

Failure cases, evidence, and limitations

Proposed verification matrix

Scenario	Method	Expected behavior
Concurrent issue	Have two staff sessions attempt to issue the same available copy.	Exactly one active loan is created; the other session receives a current availability message.
Reservation fairness	Expire a reservation and return a copy while multiple students are waiting.	The documented order selects the next eligible borrower without dropping other entries.
Renewal constraints	Attempt renewal with an expired account or another active reservation.	The rule is consistently enforced and its reason is visible.
Catalogue usability	Ask users to locate a specific edition and distinguish available from reserved copies.	Record task correctness and misunderstanding of availability labels.

Technical review

Separate titles from physical copies and preserve each loan transaction. Prevent competing issue requests with database constraints, and test reservation expiry without changing the original queue order silently.

How to evaluate the design

Measure the rate of conflicting active loans, catalogue task success, median completion time for issuing and returning a copy, and the number of unexplained reservation-order changes. Use a fixed synthetic catalogue for repeatable tests and a separate supervised usability session.

The matrix specifies intended tests, not completed test results. A reviewer should capture the environment, initial records, inputs, expected state, observed state, and any unresolved defect before interpreting the outcome. Repeated runs should use separate unique request keys where relevant and preserve failures for analysis.

Scope limitations

No interlibrary exchange, payment collection, copyrighted full-book downloads, automated disciplinary decisions, or replacement of physical stock inspection.

Check that title availability is calculated from copies and valid reservations, rather than a manually edited total that can drift from actual circulation.

> **Authorship and reuse**
>
> Prepared by Camille Navarro as an original presentation work sample. All narrative, schemas, and test scenarios are original illustrative material. There are no external client documents, employment certificates, real personal records, or fabricated citations in this case study.